Dempsey Center Client Survey Analysis (2022–2025)

1. Introduction

In this project, we analyzed client survey data from the Dempsey Center across four years (2022–2025). The goal of this project was to better understand client satisfaction, service usage, and the factors that contribute to improved life outcomes for clients.

The motivation for this analysis comes from the need to move beyond simple summary statistics and identify deeper insights that can help the Dempsey Center improve its services and better support its clients.

2. Background

The Dempsey Center is an organization that provides support services for individuals impacted by cancer, including physical, emotional, and social support programs. Each year, the Center collects survey data from clients to evaluate service effectiveness and identify areas for improvement.

Previous reports, such as the 2024 Annual Client Survey Summary, mainly focused on descriptive statistics such as satisfaction rates and service usage percentages. While these summaries provide useful information, they do not explain why certain outcomes occur or identify specific groups of clients who may need additional support.

3. Methods / Analysis

3.1 Demographics

The dataset includes demographic information such as age, gender, race, and cancer impact status (e.g., patient, survivor, caregiver). These variables were used to explore whether demographic factors influence satisfaction and outcomes.

3.2 Data Formatting

The survey data required preprocessing before analysis. Binary encoding was applied to convert Yes/No responses into numerical values (1/0). Multi-select questions, such as services used and types of support needed, were transformed into multiple binary columns. In addition, categorical variables like satisfaction levels were encoded numerically to make them suitable for statistical and machine learning methods.

3.3 Data Cleaning

Several data cleaning steps were performed to ensure consistency and usability. Missing values were handled by removing incomplete records or applying simple imputation where appropriate. Since the survey structure varied across years, column names were standardized, and data from all four years were merged into a single dataset for analysis.

3.4 Models and Analytical Methods

To analyze the data, we applied a combination of statistical and machine learning methods. Correlation analysis was used to explore relationships between variables, while a decision tree model was used to identify key predictors of satisfaction. K-means clustering (k=3) was applied to segment clients into groups based on their behavior and outcomes. In addition, chi-square tests were conducted to evaluate the statistical significance of key relationships.

4. Results

The results indicate that the Dempsey Center is performing at a high level, with around 90% of clients reporting satisfaction and over 95% indicating that their lives improved. However, deeper analysis reveals that visit frequency is the strongest predictor of satisfaction. Clients who visited more frequently reported significantly better outcomes, and this relationship was confirmed by both the decision tree model and chi-square testing ($p < 0.001$).

In contrast, demographic factors such as age, gender, and cancer were not significant predictors of satisfaction, suggesting that service quality is consistent across different client groups. Clustering analysis further identified three distinct segments, including a small group of clients with low satisfaction and minimal improvement, which would not be visible from averages alone.

Finally, trends over time show an increase in one-time visitors and a decline in high-frequency users, indicating potential challenges in client retention. A temporary rise in wait time issues in 2023 may also explain a slight drop in satisfaction during that year.

5. Discussion

The analysis provides several important insights into client outcomes and service effectiveness. Although overall satisfaction and life improvement rates are very high, these averages mask meaningful differences within the client population. In particular, the clustering results reveal a small group of clients who report low satisfaction and minimal improvement, highlighting the importance of segmentation in understanding user experience.

Among all variables, visit frequency emerged as the most important factor influencing satisfaction. Clients who engaged with services more frequently consistently reported better outcomes, suggesting that repeat engagement is critical to maximizing the impact of the Center's services. In contrast, demographic factors were not found to significantly influence satisfaction, indicating that the Center provides a consistent level of care across different client groups.

At the same time, trends over time suggest a potential decline in engagement, as the proportion of one-time visitors has increased while high-frequency usage has decreased. This pattern suggests that improving client retention may be a key opportunity for enhancing overall outcomes.

6. Limitations

This analysis has several limitations that should be considered when interpreting the results. First, the dataset is based on survey responses, which introduces self-selection bias, as individuals who choose to respond may differ from those who do not. Second, the absence of a unique client identifier prevents tracking individual behavior across years, limiting the analysis to population-level trends.

Additionally, service usage was encoded as binary variables, which does not capture the intensity or frequency of usage. This means that clients with very different levels of engagement may appear similar in the data. Finally, the dataset is highly imbalanced, with most respondents reporting high satisfaction, which may reduce the effectiveness of predictive models in identifying lower-satisfaction cases.

7. Future Work

Future work will focus on extending this analysis toward more predictive and actionable approaches. One direction is to develop models that can identify first-time visitors who are unlikely to return, allowing for targeted interventions to improve retention. Another important step is to introduce unique client identifiers, which would enable longitudinal analysis of individual outcomes over time.

In addition, experimental approaches such as A/B testing could be used to evaluate the effectiveness of follow-up programs and other engagement strategies. Expanding the dataset to include more detailed behavioral and operational information, such as service frequency and scheduling data, would further improve the ability to understand and address client needs.